

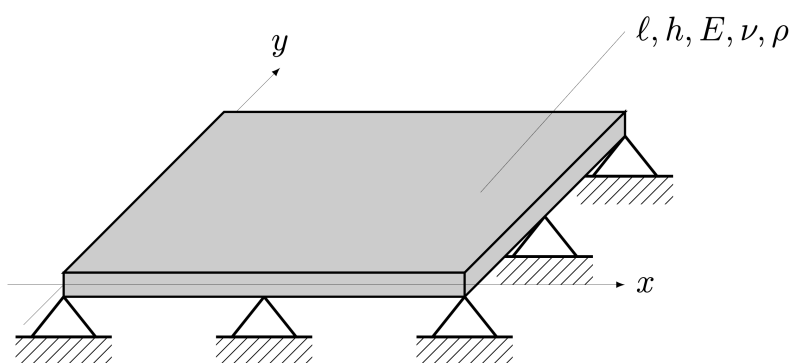


ME-473: DYNAMIC FINITE ELEMENT ANALYSIS OF STRUCTURES

Mini-project 3 report - Natural frequencies and mode shapes of a square plate

GROUP F

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1 Introduction

The goal of this mini-project is to study the natural frequencies of a square, isotropic, thin plate, simply supported along all four edges. The first part of the report presents the modelling of the plate using AMC plate bending elements and studies the effect of increasing the mesh refinement to check the convergence of the results with the analytical solution. In the second part of the report, CR plate bending elements are considered and the span over thickness ratio is modified. For different ratios, the results of the simulation are compared with the solutions of both the thin plate and thick plate analytical formulations. The software used to code and run the simulation is MATLAB.

2 Methodology

In this case, a modal analysis will be conducted to examine the transversal vibrations of a thin plate. The plate in question is defined by the following parameters:

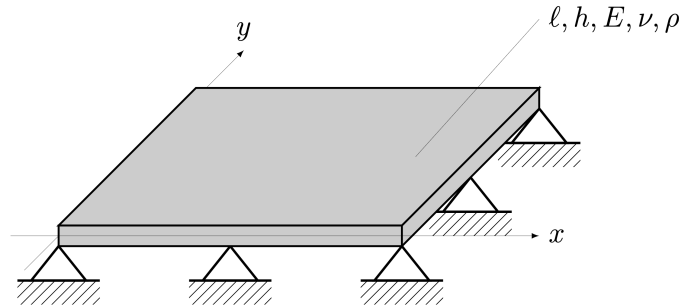


Figure 1: Schematic of the simply supported thin plate.

- Edge length: $l = 0.5$ m,
- Thickness: $h = 0.005$ m,
- Young's modulus: $E = 210$ GPa,
- Poisson's ratio: $\nu = 0.3$,
- Density: $\rho = 7850$ kg/m³.

The plate is simply supported, this implies no displacement and no torque (which allows free rotation along the edge) at the edge. Mathematically speaking, the boundary conditions are $u_3 = 0$ and $\mathbf{M} = 0 \forall t$ in $\partial\Omega$, where $\Omega = [0, l] \times [0, l]$. The initial conditions are zero displacement and zero velocity, thus $u_3(t = 0) = 0$ and $\dot{u}_3(t = 0) = 0$ in Ω . Furthermore, as only a modal analysis will be considered, there is no external force on the system, so $f_3 = 0$.

2.1 Assumptions

The assumptions made along this project are the following:

- All properties are homogeneous and isotropic,
- We assume small deformation, the displacement u_3 is considered small compared to h ,
- The thickness h is constant, much smaller than the length l and remains inextensible,
- The system must remain in the linear regime,
- Plane stress state is assumed,

- The plate is considered as thin, therefore the rectilinearity of the normals can be assumed,
- The displacement vector is given as $\mathbf{u} = [-z \frac{\partial u_3}{\partial x}, z \frac{\partial u_3}{\partial y}, u_3]^T$, so the transversal displacement u_3 is the only independent variable,
- The gravitational forces are neglected.

2.2 Equations of motion

Thanks to the assumptions made, we can state the strong form of the Kirchhoff-Love plate bending theory [1] with the given boundary and initial conditions:

$$\begin{cases} \frac{h^3}{12} \nabla_k^T \mathbf{C} \nabla_k u_3 + \rho h \ddot{u}_3 = 0 & , \text{ on } \Omega \times]0, T[\\ u_3 = 0 \text{ and } \mathbf{M} = 0 & , \text{ in } \partial\Omega \times]0, T[\\ u_3(\cdot, t = 0) = 0 \text{ and } \dot{u}_3(\cdot, t = 0) = 0 & , \text{ in } \Omega \end{cases}$$

with

$$\mathbf{C} = \frac{E}{(1 - \nu^2)} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

Then, we can extract the weak form, which is also directly derived from the lecture notes [2]:

$$\boxed{\frac{h^3}{12} \int_{\Omega} (\nabla_k \delta u_3)^T \mathbf{C} \nabla_k u_3 d\Omega + \int_{\Omega} (\delta u_3)^T \rho h \ddot{u}_3 d\Omega = 0}$$

where $u_3 \in \mathcal{U} = \{u_3(\cdot, t) \in H^2(\Omega) \mid u_3 = 0 \text{ in } \partial\Omega \times]0, T[\}$ and $\delta u_3 \in \mathcal{V} = \{\delta u_3 \in H^2(\Omega) \mid \delta u_3 = 0 \text{ on } \partial\Omega\}$.

2.3 Displacement approximation

We can define the approximation of displacement and the virtual displacement:

$$u_3^h(x, y, t) = \mathbf{H}(x, y) \mathbf{q}(t)$$

$$\delta u_3(x, y, t) = \mathbf{H}(x, y) \delta \mathbf{q}(t)$$

Using these precedent relations into the weak form, we obtain the following.

$$\begin{aligned} \frac{h^3}{12} \int_{\Omega} (\nabla_k \mathbf{H} \delta \mathbf{q})^T \mathbf{C} (\nabla_k \mathbf{H} \mathbf{q}) d\Omega + \int_{\Omega} (\mathbf{H} \delta \mathbf{q})^T \rho h (\mathbf{H} \ddot{\mathbf{q}}) d\Omega &= 0 \\ \frac{h^3}{12} \int_{\Omega} (\delta \mathbf{q})^T (\nabla_k \mathbf{H})^T \mathbf{C} (\nabla_k \mathbf{H}) \mathbf{q} d\Omega + \int_{\Omega} (\delta \mathbf{q})^T \mathbf{H}^T \rho h \mathbf{H} \ddot{\mathbf{q}} d\Omega &= 0 \\ (\delta \mathbf{q})^T \left[\frac{h^3}{12} \int_{\Omega} (\nabla_k \mathbf{H})^T \mathbf{C} (\nabla_k \mathbf{H}) d\Omega \mathbf{q} + \int_{\Omega} \mathbf{H}^T \rho h \mathbf{H} d\Omega \ddot{\mathbf{q}} \right] &= 0 \end{aligned}$$

As $\delta u_3 \in \mathcal{U}$, we can deduce:

$$\frac{h^3}{12} \int_{\Omega} (\nabla_k \mathbf{H})^T \mathbf{C} (\nabla_k \mathbf{H}) d\Omega \mathbf{q} + \int_{\Omega} \mathbf{H}^T \rho h \mathbf{H} d\Omega \ddot{\mathbf{q}}$$

Then, we can rewrite it in the matrix form as following:

$$\mathbf{K} \mathbf{q} + \mathbf{M} \ddot{\mathbf{q}} = 0$$

where we can define the stiffness matrix $\mathbf{K}$ and mass matrix $\mathbf{M}$:

$$\mathbf{K} = \frac{h^3}{12} \int_{\Omega} (\nabla_k \mathbf{H})^T \mathbf{C} (\nabla_k \mathbf{H}) d\Omega$$

$$\mathbf{M} = \int_{\Omega} \mathbf{H}^T \rho h \mathbf{H} d\Omega$$

As we are going to use local shape functions for AMC and CR elements, we define:

$$\mathbf{K} = \mathbf{A} \sum_{e=1}^m {}^e \mathbf{K}$$

$$\mathbf{M} = \mathbf{A} \sum_{e=1}^m {}^e \mathbf{M}$$

where $\mathbf{A}$ is the assembly operator and where ${}^e \mathbf{K}$ and ${}^e \mathbf{M}$ are defined as following:

$${}^e \mathbf{K} = \frac{h^3}{12} \iint_{\xi_1, \xi_2} (\nabla_k {}^a \mathbf{H})^T \mathbf{C} (\nabla_k {}^a \mathbf{H}) {}^a j d\xi_1 d\xi_2$$

$${}^e \mathbf{M} = \iint_{\xi_1, \xi_2} {}^a \mathbf{H}^T \rho h {}^a \mathbf{H} {}^a j d\xi_1 d\xi_2$$

Finally, we are only interested in the modal analysis; thus we need to solve the following equations in order to compute the natural frequencies:

$$\det(\mathbf{K} - \omega_n^2 \mathbf{M}) = 0$$

where ω_n are the natural angular frequencies of the system. From this result, the eigenvectors, which will represent the mode shapes, can be computed. The normalization used was the predefined from MATLAB, so that the 2-norm of each eigenvector is 1 [3].

2.4 AMC element shape functions

We can define the local shape functions:

$${}^a \mathbf{h}_i(\xi_1, \xi_2) = \begin{bmatrix} \frac{1}{8}(1 + \xi_1^i \xi_1)(1 + \xi_2^i \xi_2)(2 + \xi_1^i \xi_1 + \xi_2^i \xi_2 - \xi_1^2 - \xi_2^2) \\ \frac{e l}{16}(1 + \xi_1^i \xi_1)(\xi_2^i + \xi_2)(\xi_2^2 - 1) \\ -\frac{e l}{16}(1 + \xi_2^i \xi_2)(\xi_1^i + \xi_1)(\xi_1^2 - 1) \end{bmatrix}^T$$

$${}^a \mathbf{H} = [{}^a \mathbf{h}_1 \quad {}^a \mathbf{h}_2 \quad {}^a \mathbf{h}_3 \quad {}^a \mathbf{h}_4]$$

where (ξ_1^i, ξ_2^i) are the local coordinates of node i , and ${}^a \mathbf{H}$ the shape functions matrix of size 1x12. From this, we can compute:

$${}^a \mathbf{B} = \nabla_k {}^a \mathbf{H} = \begin{bmatrix} \left[\begin{array}{c} \frac{4}{e l^2} \partial_{\xi_1}^2 \\ \frac{4}{e l^2} \partial_{\xi_2}^2 \\ \frac{8}{e l^2} \partial_{\xi_1 \xi_2}^2 \end{array} \right] {}^a \mathbf{h}_1 & \left[\begin{array}{c} \frac{4}{e l^2} \partial_{\xi_1}^2 \\ \frac{4}{e l^2} \partial_{\xi_2}^2 \\ \frac{8}{e l^2} \partial_{\xi_1 \xi_2}^2 \end{array} \right] {}^a \mathbf{h}_2 & \left[\begin{array}{c} \frac{4}{e l^2} \partial_{\xi_1}^2 \\ \frac{4}{e l^2} \partial_{\xi_2}^2 \\ \frac{8}{e l^2} \partial_{\xi_1 \xi_2}^2 \end{array} \right] {}^a \mathbf{h}_3 & \left[\begin{array}{c} \frac{4}{e l^2} \partial_{\xi_1}^2 \\ \frac{4}{e l^2} \partial_{\xi_2}^2 \\ \frac{8}{e l^2} \partial_{\xi_1 \xi_2}^2 \end{array} \right] {}^a \mathbf{h}_4 \end{bmatrix}$$

This type of elements will be used in the first part of the project, when increasing the number of elements and computing its convergence.

2.5 CR element shape functions

Instead of using the AMC shape functions defined above, we will be using the following ones:

$${}^a\mathbf{h}_i(\xi_1, \xi_2) = \begin{bmatrix} \frac{1}{16}(-\xi_1^i \xi_1^3 + 3\xi_1^i \xi_1 + 2)(-\xi_2^i \xi_2^3 + 3\xi_2^i \xi_2 + 2) \\ \frac{el}{16}(-\xi_1^i \xi_1^3 + 3\xi_1^i \xi_1 + 2)(\xi_2^3 + \xi_2^i \xi_2^2 - \xi_2 - \xi_2^i) \\ \frac{el}{16}(\xi_1^3 + \xi_1^i \xi_1^2 - \xi_1 - \xi_1^i)(-\xi_2^i \xi_2^3 + 3\xi_2^i \xi_2 + 2) \\ \frac{el^2}{16}(\xi_1^3 + \xi_1^i \xi_1^2 - \xi_1 - \xi_1^i)(\xi_2^3 + \xi_2^i \xi_2^2 - \xi_2 - \xi_2^i) \end{bmatrix}^T$$

$${}^a\mathbf{H} = [{}^a\mathbf{h}_1 \quad {}^a\mathbf{h}_2 \quad {}^a\mathbf{h}_3 \quad {}^a\mathbf{h}_4]$$

The shape function ${}^a\mathbf{H}$ has now size 1x16. This kind of elements will be used in the second part of the project, when increasing the span over thickness ratio and comparing it to the thick plate theory.

2.6 Natural frequency analytical solution

To compare the numerical solutions, we'll be using the analytical solution, which is:

$$f_{m,n} = \frac{\pi}{2} \frac{n^2 + m^2}{l^2} \sqrt{\frac{D}{\rho h}}$$

where n and m are the number of half-waves in x and y directions, l is the length of the square plate, h is the thickness and D is the flexural rigidity of the plate and which is defined as:

$$D = \frac{Eh^3}{12(1 - \nu^2)}$$

Then, when increasing the span over thickness ratio and noticing the evolution of the natural frequency, we will compare the numerical solutions to the thick plate theory. This will be made using the premade *MATLAB* function `computeThickFrequency`.

3 Results and Discussion

This section presents and analyses the numerical results obtained from the finite element simulations. Two main aspects are studied: the influence of mesh refinement using the AMC element formulation, and the impact of the length-to-thickness ratio using the CR elements. The computed natural frequencies are compared with analytical solutions for both thin and thick plate theories to assess the accuracy and convergence behaviour of the numerical models.

3.1 Effect of Mesh Refinement (AMC)

Mode	Analytical Solution	Numerical Results using AMC			
		2x2	4x4	8x8	16x16
$f_{1,1}$ [Hz]	98.34	89.78	95.41	97.54	98.14
$f_{1,2}$ [Hz]	245.86	253.65	236.18	242.82	245.05
$f_{2,1}$ [Hz]	245.86	253.65	236.18	242.82	245.05
$f_{2,2}$ [Hz]	393.37	540.87	359.14	381.63	390.18

Table 1: Summary table of the natural frequencies obtained, all values are given in **Hz**.

Table 1 presents the natural frequencies of a square plate for the first four vibration modes, comparing the analytical solution with numerical results obtained using the AMC elements at different mesh densities. This method provides increasingly accurate frequency as the mesh is refined. For the fundamental mode $f_{1,1}$, the numerical results get very close to the analytical value even with a relatively coarse mesh. For the higher modes, the difference is larger when the mesh is coarse, but the accuracy improves a lot with a 16x16 mesh.

From these results, we can calculate the relative error between the numerical results and the analytical solution. The relative error is then defined as:

$$\delta_{i,j}^n = \frac{f_{i,j}^n - f_{i,j}^{exact}}{f_{i,j}^{exact}}$$

Mode	Error of Numerical Results [-]			
	2x2	4x4	8x8	16x16
$\delta_{1,1}^n$ [%]	-8.703	-2.986	-0.812	-0.207
$\delta_{1,2}^n$ [%]	3.169	-3.938	-1.236	-0.328
$\delta_{2,1}^n$ [%]	3.169	-3.938	-1.236	-0.328
$\delta_{2,2}^n$ [%]	37.496	-8.703	-2.986	-0.812

Table 2: Summary table of the natural frequencies errors computed, all values are given in %.

Table 2 shows the relative errors (in percent) between the AMC numerical results and the analytical solution for each mode and mesh density. The results confirm the trend observed previously: the error decreases significantly as the mesh becomes finer. The largest errors appear in the higher modes at coarse resolutions, but these errors drop below 1% at 16x16 mesh, indicating good convergence of the method.

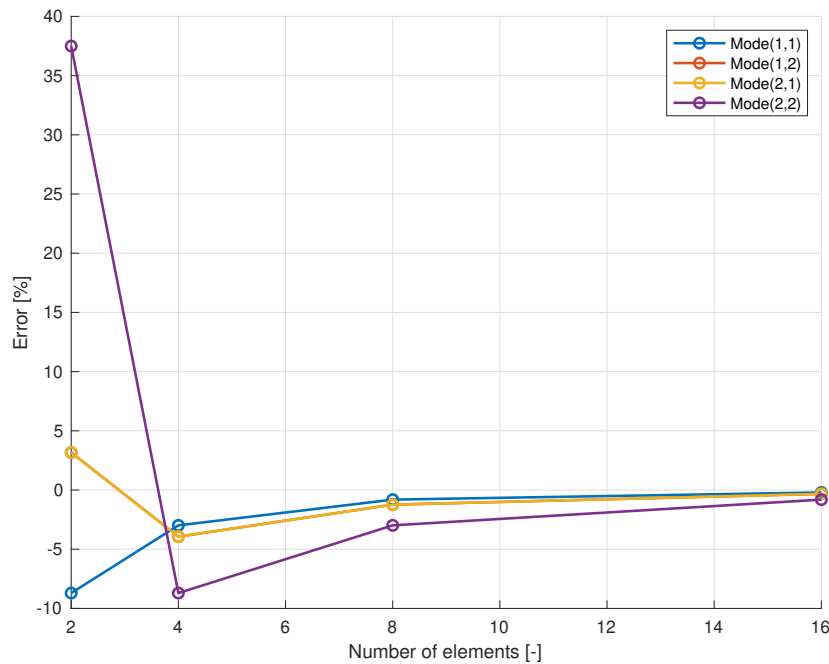


Figure 2: Error between the numerical solution using AMC elements and the analytical solution with increasing number of elements.

Figure 2 presents the relative errors from Table 2, plotted against increasing mesh density for the first four vibration modes. This visualization confirms the convergence of the AMC method: as the number of elements increases, the numerical results approach the analytical solution. The plot clearly illustrates the need for finer meshes to accurately capture complex mode shapes. We also observe, as discussed in class, that the relative error is negative, meaning that the use of AMC elements tends to underestimate the natural frequencies and thus provides a lowerbound on the estimation of the exact natural frequencies.

All the mode shapes are available in the Appendices A, B, C and D. However, certain results are particularly noteworthy and merit closer examination.

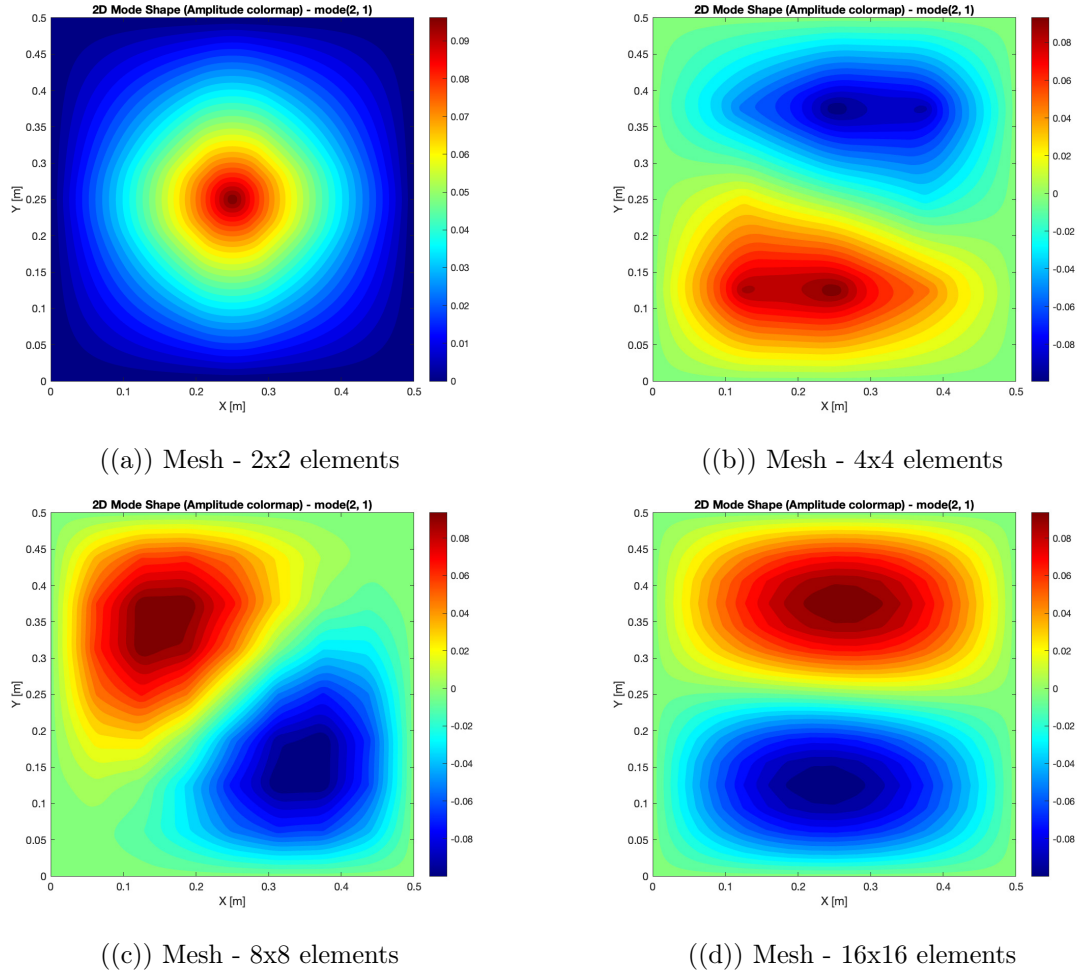


Figure 3: Second mode shape (2,1) with increasing number of elements.

Figure 3 shows the second mode (2,1) computed using AMC elements with different mesh refinements. With only 2x2 elements, the mode shape is poorly resolved and deviates from the expected analytical pattern. However, with 4x4 elements, the shape becomes much more consistent with theoretical expectations.

3.2 Effect of the thickness ratio (CR)

To compare the numerical results and the analytical solution for the thin plate with the analytical solution for the thick plate, we use this relation:

$$f_{thick} \approx 2'547.77\pi^2 h \sqrt{\frac{4'155'234.46}{393'216\pi^2 h^2 + 172'032}}$$

Note: Here, we show the formula for a plate with the geometry and material properties

specified at the beginning of the report. For details about the computation, please refer to the `computeThickFrequency` function.

Ratio l/h	Thin plate [Hz]	Thick plate [Hz]	CR [Hz]
10	983.43	956.82	998.68
20	491.71	488.28	499.34
30	327.81	326.79	332.89
40	245.86	245.43	249.67
50	196.69	196.46	199.74
60	163.90	163.78	166.45
70	140.49	140.41	142.67
80	122.93	122.87	124.83
90	109.27	109.23	110.96
100	98.34	98.32	99.87

Table 3: Comparison of frequencies for different l/h ratios using 8x8 elements.

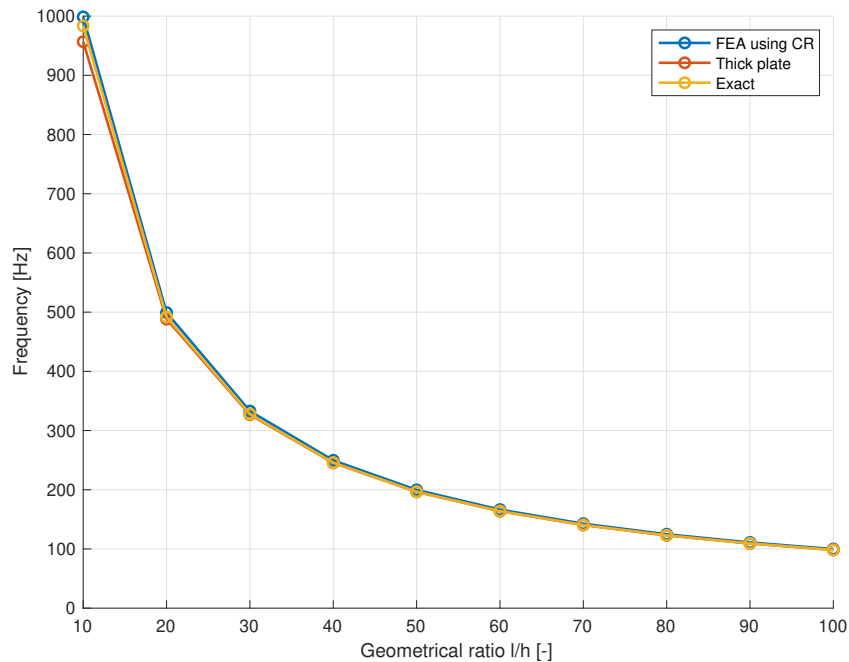


Figure 4: Comparison of the 1st mode frequency (1,1) between the numerical solution using CR elements, the exact analytical solution and the thick plate solution; watch out, the thickness increases following the axis from right to left

From figure 4 and table 3 we observe that the thin plate formulation and the thick one converge as the span/thickness ratio increases. However, for small ratios, the thin plate assumptions are not valid and the thin model diverges from the thick one. The model with CR bending element follow the evolution of the thin plate analytical model and overestimates the natural frequency, providing an upperbound.

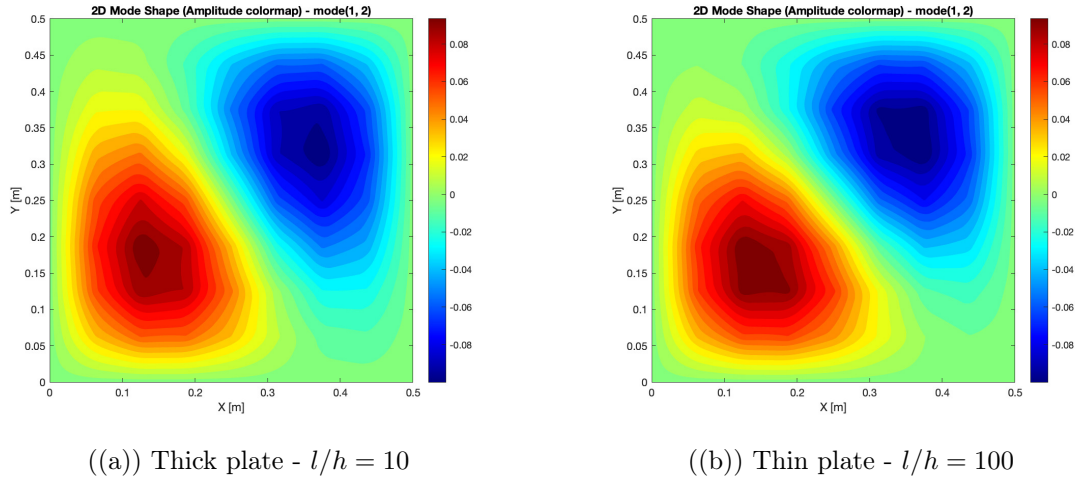


Figure 5: Second mode shape (1,2) with increasing span-to-thickness ratio.

Figure 5 illustrates the second mode shape (1,2) of a square plate for two different span-to-thickness ratios, $l/h = 10$ and $l/h = 100$, using the CR element formulation. The color maps represent the transverse displacement u_3 across the plate surface.

3.3 Discussion

The results show that refining the mesh significantly improves the accuracy of simulations using AMC elements. As we increase the number of elements, the computed frequencies become closer to the exact analytical solutions. For instance, the first vibration mode $f_{1,1}$ is already predicted quite well with a 4x4 mesh. However, the higher vibration modes, which have more complex shapes, require finer meshes to be properly captured. This improvement is confirmed by the error values: for all the computed modes, the relative error decreases as the mesh becomes finer. When we reach a 16x16 mesh, the error falls below 1%, which shows that the AMC method becomes highly accurate with enough refinement. This means that AMC elements can effectively capture the behavior of vibrating plates, as long as we use a sufficiently dense mesh.

In addition, while we were able to compute accurate values for higher modes such as (1,2), (2,1), and (2,2), there is still a limitation in how the displacements are represented in our model. Each node currently has only one degree of freedom for the displacement u_3 , which restricts the way we can describe complex vibration shapes. This is especially clear in Figure 3, where we show the (2,1) mode shape across all meshes size. With only a 2x2 mesh, the mode shape is poorly represented, mainly because only one node is allowed to displace in the z direction.

Moreover, as the mesh becomes finer, the mode shapes not only become more accurate but also visually sharper, more defined and bigger. With a coarse mesh, the mode shapes appear smooth and rounded, lacking the characteristic nodal lines and spatial variation seen in analytical solutions. As shown in Appendix D, the (2,2) mode shape transitions from 4 circular central regions at 4x4 elements to clearly separated rectangular lobes at 16x16 elements. This evolution illustrates how a denser mesh enhances the resolution of the computed displacement field, making the pattern more angular and aligned with the geometry of the square plate.

In the second part of the study, we focused on how the plate's thickness affects the simulation results using CR elements. As the plate becomes thicker (lower l/h ratio), the computed natural frequencies follow the thin plate theory and shift away from the thick plate theory values. The CR formulation agrees with the thin plate theory and those values are exact when the plate is thin. However, as the thickness gradually increases, the thin plate theory becomes less and less valid and the thick plate formulation has to be chosen instead.

While the mode shapes for both cases generally follow the same global patterns (e.g., number and orientation of nodal lines), noticeable differences appear in their overall deformation

characteristics. For the thinner plate ($l/h = 100$), the mode shapes display a highly symmetric deformation, with clear separation along the diagonals, forming distinct triangular patterns near the corners. In contrast, for the thicker plate ($l/h = 10$), these triangular regions become noticeably distorted: they are squashed in one direction and stretched in the other, taking on more trapezoidal shapes. This asymmetry reflects the influence of transverse shear deformation, which becomes significant in thicker plates and is not accounted in Kirchhoff-Love plate theory. These observations underscore the necessity of using shear-deformable plate models when analyzing plates with lower l/h ratios to more accurately capture their modal behavior.

One interesting thing we can observe from the frequency error table is the sign of the errors: the AMC results always give lower frequencies than the exact ones. This is because AMC elements are non-conforming, meaning their shape functions do not fully enforce continuity across element boundaries. This leads to a slight underestimation of the system's stiffness, and therefore, lower predicted frequencies. On the other hand, CR elements are conforming, they satisfy the continuity conditions between elements. As a result, the structure behaves slightly stiffer, and the predicted frequencies tend to be higher than the exact values. This explains why AMC errors are always negative and CR errors are always positive.

4 Conclusion

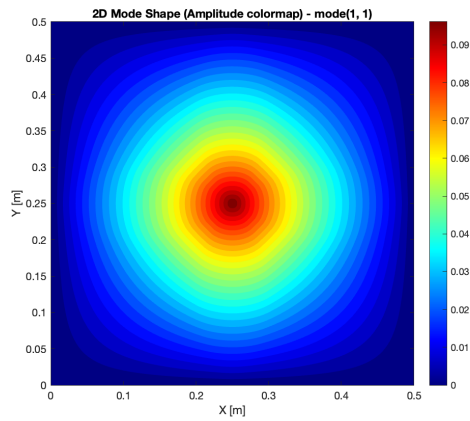
This mini-project dived into the study of a vibrating plate, simply supported along its edges. The plate was modelled using both AMC and CR plate bending elements. Using the AMC elements, the sensitivity of the simulations to the mesh refinement and the result convergence were studied. Finally, the sensitivity of the CR elements to the span/thickness ratio was studied and compared with the thin and thick plate theory.

The results highlighted the improvement of the simulations when increasing the mesh refinement and the divergence of the simulation with the exact model as the thickness increases. These observations showed the importance of selecting a mesh size that corresponds to the accuracy and to the complexity of the modes we want to simulate. Moreover, the geometrical ratio between the length of the element and its thickness have to be taken into account when choosing between the thin or thick plate formulation for our simulations.

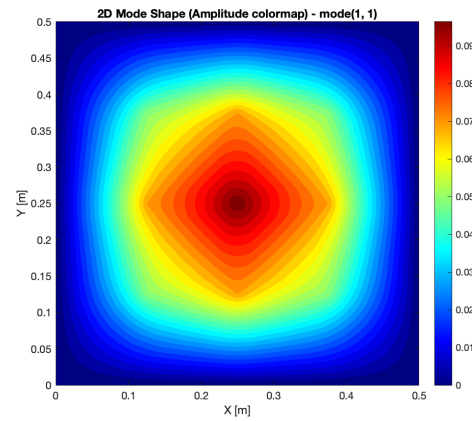
References

- [1] Thomas Gmür. *Dynamique des structures*. Presses polytechniques et universitaires romandes, 2014.
- [2] Stefano F. Burzio. *Lecture Notes: ME-473: Dynamic Finite Element Analysis of Structures*. Spring semester, EPFL. 2025.
- [3] MATLAB. *eig - Eigenvalues and eigenvectors*. Accessed April 2025. URL: <https://ch.mathworks.com/help/matlab/ref/eig.html>.

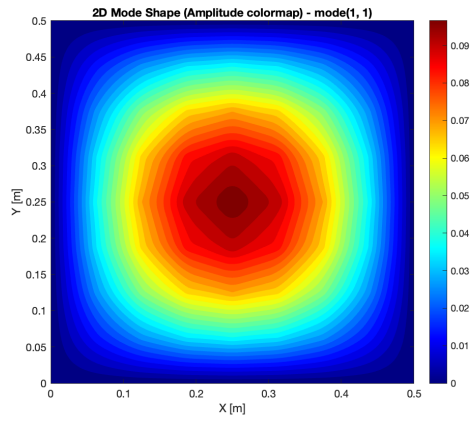
A Mode (1,1) - Varying Mesh size



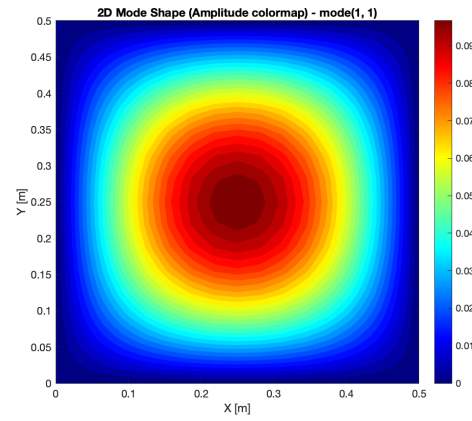
((a)) Mesh - 2x2 elements



((b)) Mesh - 4x4 elements



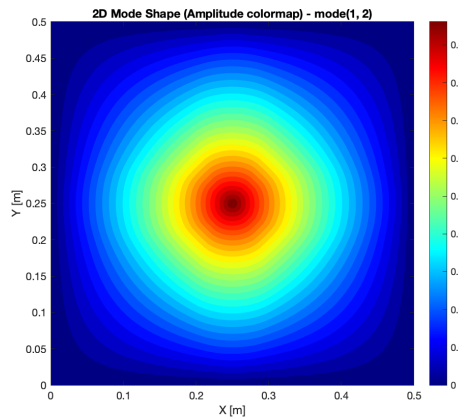
((c)) Mesh - 8x8 elements



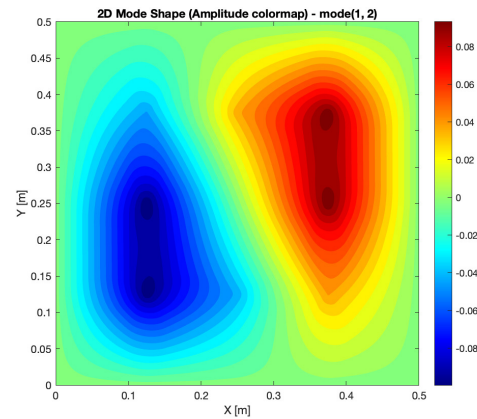
((d)) Mesh - 16x16 elements

Figure 6: FEM Analysis mode (1,1)

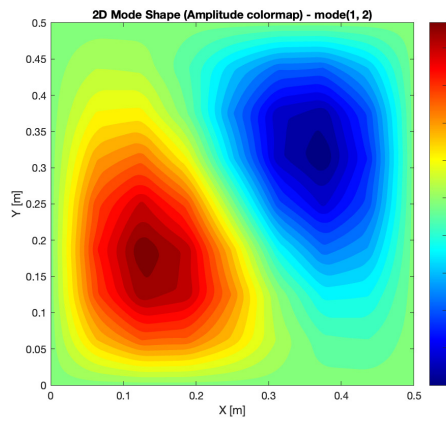
B Mode (1,2) - Varying Mesh size



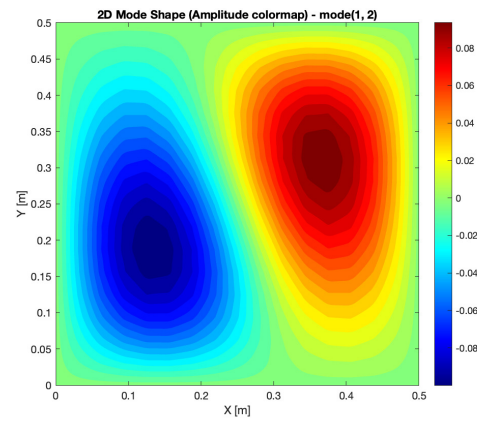
((a)) Mesh - 2x2 elements



((b)) Mesh - 4x4 elements



((c)) Mesh - 8x8 elements



((d)) Mesh - 16x16 elements

Figure 7: FEM Analysis mode (1,2)

C Mode (2,1) - Varying Mesh size

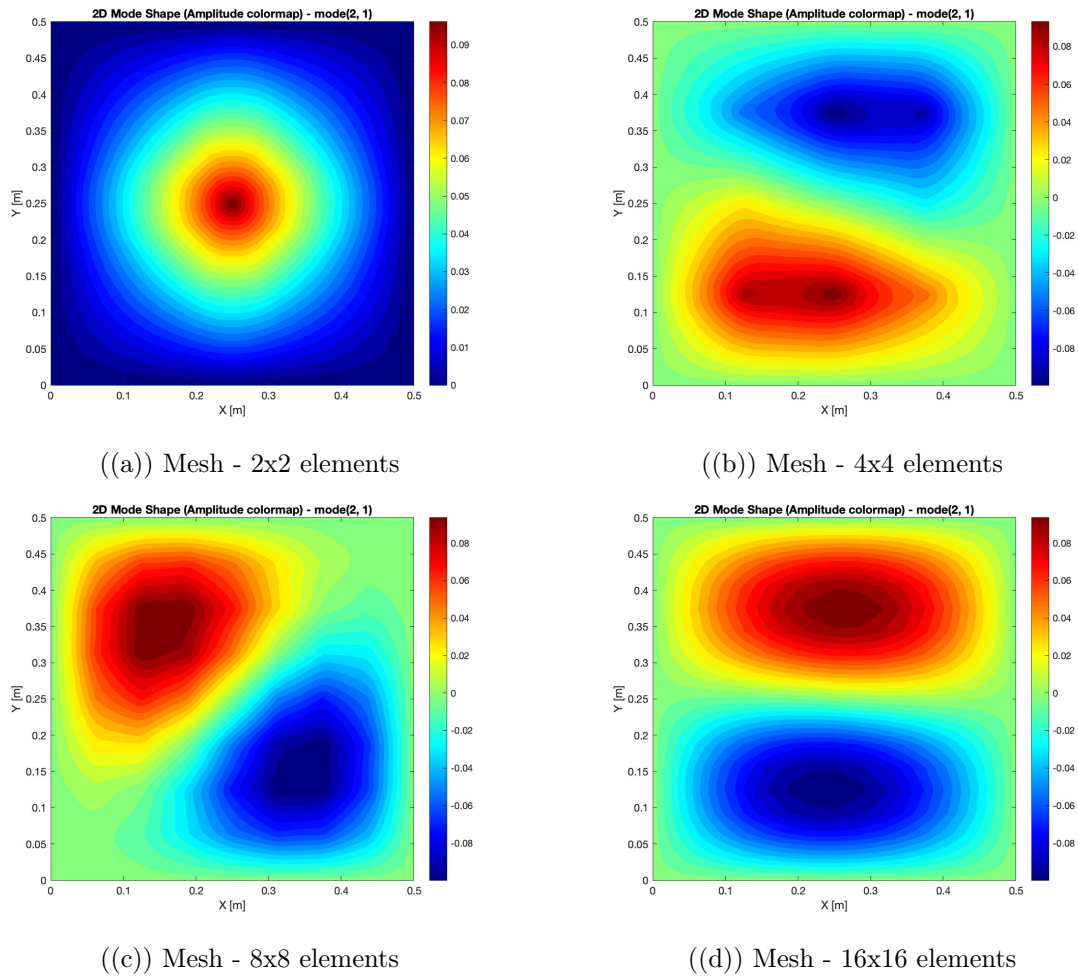
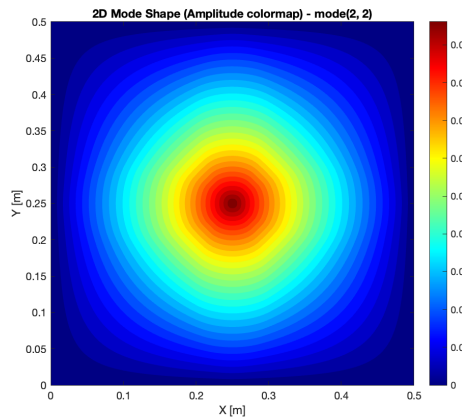
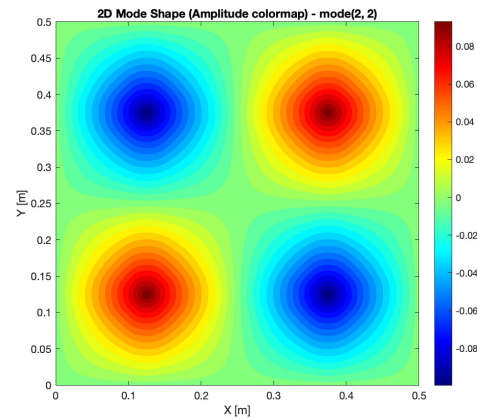


Figure 8: FEM Analysis mode (2,1)

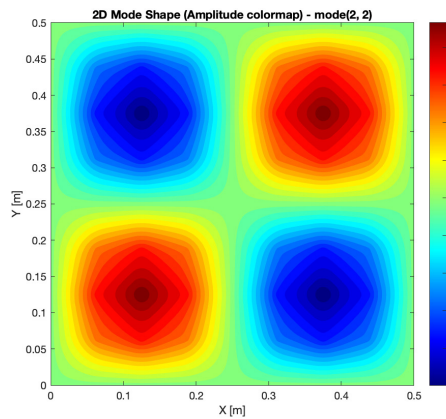
D Mode (2,2) - Varying Mesh size



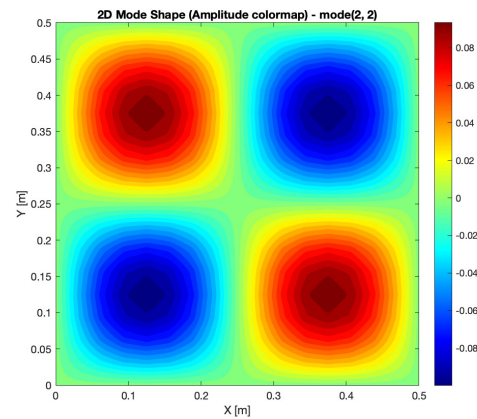
((a)) Mesh - 2x2 elements



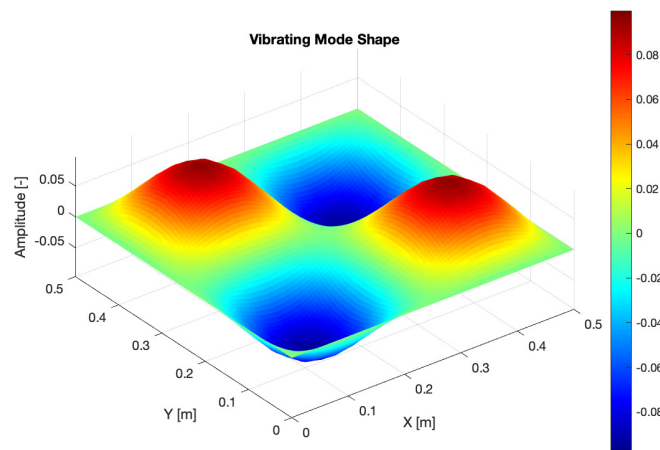
((b)) Mesh - 4x4 elements



((c)) Mesh - 8x8 elements



((d)) Mesh - 16x16 elements



((e)) 3D view of mesh - 16x16 elements

Figure 9: FEM Analysis mode (2,2)